Abhishek Maiti

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EDUCATION

IIIT Delhi, *New Delhi, India*
Bachelor of Technology in Computer Science and Engineering (Honors) with minor in Economics *GPA: 8.56 / 10.00*

WORK EXPERIENCE

Cerebras, *Abu Dhabi, UAE*
Member of Technical Staff 2 *April. 2025 – Present.*

- Leading RL and Evaluations on the Jais team. Increasing tool calling, multiturn and instruction following capabilities across 4B dense to 400B MoE decoder only transformer based model scales. Designing efficient evaluation frameworks for both upstream and downstream benchmarking.
- Core contributor to Jais 2 8B/70B (post-training); SOTA results on Ar & En benchmarks via eval unification, distributed RLHF, and tool-calling/Long Context research.
- `[Evals]` Built unified eval/RL infra (3× throughput).
- `[RL]` Enabled multi-node RLHF/GRPO on 8B/70B using LLM as a judge rewards. (Results: **+8pp** GSM8K, **+4pp** Arabic-IFEval, **+6pp** Vicuna).
- `[Tool Calling]` Achieved BFCL-v3 **+12pp** and HotPotQnA pass@k **+17pp** via synthetic & Long Context RL.
- `[Language Adaptation]` Designed efficient language adaptation training curriculum to adapt non-Arabic models to Arabic (both post trained and pretrained models), increasing model performance by **+4pp** averaged across benchmarks, compared to natively pretrained dense models.

EarnIn, *Bangalore, India*
Senior Machine Learning Engineer *March. 2024 – April. 2025*

- Developed and implemented customer risk profiling system using financial transactions and GPS data, leveraging LLMs and feature engineering to achieve **15%** increase in profile accuracy and **10%** year-over-year margin improvement.
- Pretrained and fine-tuned LLMs on bank transactions for generating embeddings for downstream payroll and earnings related tasks. Achieved **15-20%** metric boost across key business areas.
- Built a automated customer ticket resolution Agent, cutting time to resolution by almost 2x.
- Designed efficient algorithms to predict paycheck dates, hourly rates and exact date of job change for gig-workers in the US.

LinkedIn AI, *Bangalore, India*
Applied Research Engineer *April. 2022 – February. 2024*

- Enhancing image content moderation with the Trust AI team using multi-modal strategies across various policies, saving upwards of **2.5%** of total weekly policy violating views.
- Streamlining models using a first-of-a-kind, in-house multitasking framework to lessen maintenance without compromising performance, reducing classifier count by **2x**.
- Designing and implementing an Auto-Retraining framework to minimize manual maintenance, saving **10 developer-days** per classifier for over 60 classifiers.
- Designing (with patent under review) a novel method to procure finegrained moderation feedback (image region level) from users with less than 3 clicks for the user.

The Wadhwani Institute for Artificial Intelligence, *Mumbai, India*
Associate Machine Learning Scientist - I *June. 2020 – April. 2022*

- Developing 3D human reconstruction system for detecting low birth weight in babies through monocular videos, improving keypoint estimation accuracy with OKS increase from **0.1 to 0.6**. [Problem Statement]
- Enhanced data pipeline efficiency with **3x** faster model iteration and **4x** reduction in domain gap, while improving data annotation quality for **2x** decrease in MAE.
- Developed an automated pipeline for 3D mesh registration with scans, using a derivative of SMPL model for India babies, reducing scan registration time by almost 3x compared to it being done manually. Inferring accurate pose and shape vector estimation from noisy 3D hospital scans.

PUBLICATIONS & PATENTS

- *Technical Report* **A. Maiti** (**Core Post Training Team**) ”Jais 2: A Family of Arabic-Centric Open Large Language Models”, [Report]
- *Paper* **A. Maiti***, V. Udandarao*, S. Vyalla*, D. Srivatsav*, Y. Yin, R. Shah, ”COBRA: Contrastive Bi-Modal Representation Learning”, 2nd TUSION Workshop held in conjunction with International Joint Conference on Artificial Intelligence (IJCAI-PRICAI) - 2020 [Paper Link]
- *Paper* **A. Maiti***, D. Mohapatra*, S. Bhatia and T. Chakraborty, ”Go Wide, Go Deep: Quantifying the Impact of Scientific Papers Through Influence Dispersion Trees,” 2019 ACM/IEEE Joint Conference on Digital Libraries (JCDL), Champaign, IL, USA, 2019.[**Best Student Paper Award**] [Paper Link]
- *Filed Patent* (*960106-US-NP*) S. Dey, **A. Maiti**, S. Bose, C. Voelkel, S. Marvaniya, V. Gupta, V. Jain ”Machine Learning-assisted User Reporting And Moderation Of Multimedia Content”

- *Paper*: J. Shin, **A. Maiti**, Y. Zou, J. Choi, ”Multi-Teacher Invariance Distillation for Domain-Generalized Action Recognition”, 27th International Conference on Pattern Recognition (ICPR), Kolkata, India, 2024. [Paper Link]
- *Paper*: NurtureNet: ”A Multi-task Video-based Approach for Newborn Anthropometry”. Computer Vision and Pattern Recognition Workshop (2024). [**Best Paper Award**] [Paper Link]

PROJECTS

Independent Consultation, *Worldwide*
Research Engineer

- Locally hosted Deep-Research tool on datasheets for employees in semi-conductor industry, using RL and cold-start SFT. Improving F1 score by **+5.2pp** over Gemini 2.5 pro, using Qwen3-8B model.
- Finetuning LLMs for local contextual job training for frontline workers and local administrative staff in rural India.
- Created an AI sales assistant, increasing sales outreach by 10x, with personalized emails using fine-tuned LLMs, costing $0.1 per email.
- Created a 10x faster real-time talking head animation pipeline robust against occlusions and extreme poses using pose estimation, 2D/3D landmarks.
- Finetuned custom stable diffusion, CLIP models – performant for super-resolution, inpainting, outpainting tasks.
- Developed a 30s single-view 3D human reconstruction module, with real time mesh generation from user supplied photos or measurements.
- Developed sensor-free Golf pose estimation, using 2D and 3D landmarks, finetuned on noisy images.

Personal Projects, *Maintainer*

- **Stable Fashion** First prompt based virtual try on api, which uses 2D images, various segmentation modules and finetuned stable diffusion checkpoint. Recognized as the Huggingface Spaces of the Week. 89 ★on .
- **nano-vllm** Minimal and educational implementation of vLLM engine. It comes with an educational mode to teach the inner workings of a high througput inference engine. 177 ★on .
- **RAT** A Python package for tracing attention paths backward through transformer models, with interactive D3.js Sankey visualization. 4 ★on .
- `mlresearchengineer.substack.com`: Substack blog with niche ML experiments. 100 subscribers.
- `workatafrontierlab.com`: ML interview preperation site with 1000+ users.

Frameworks Used

- **Libraries**: Transformers (Huggingface), TRL (Huggingface), veRL, PyTorch, LangChain, LlamaIndex, vLLM, SGLang, Tensorflow.
- **Languages**: Python, Java, C++